Super-Resolution by playing with Deep Image Prior

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We choose super resolution as the task due to it's complexity and with works still being published as discussed in the class.

> *We bent the losses, warped the space,*
> *Tested transformers in the race,*
> *Pushed priors through the gradient maze—*
> *To see what truly makes them blaze.*
>
> *And in that quest, one truth stood tall:*
> *You don't need million weights at all.*
> *A slim **ConvNeXt** stepped in the light,*
> *Stripped down in size, yet twice as bright—*
> *Matching the classic DIP stride for stride,*
> *With tiny parameters on its side!*

Abstract

Deep Image Prior (DIP) demonstrated that an untrained convolutional network acts as an implicit natural image prior, enabling zero-shot restoration without external training data. In this work, we systematically investigate the boundaries of this inductive bias for 4x single-image super-resolution across three dimensions: **architectural priors**, **objective and geometric regularizers**, and **learning paradigms**.

First, by evaluating diverse network families (U-Net, ResNet, Swin Transformer, ConvNeXt, and Depthwise-Separable Skip), we reveal that DIP's representational power is governed by **multi-scale hierarchical topology** rather than raw parameter volume: replacing standard convolutions with depthwise-separable convolutions slashes parameters by 84% (from 2.22M down to 346K) while preserving over 70% of DIP's performance gain on the Set14 benchmark. Furthermore, we modernize the architecture with ConvNeXt-Skip dB. Second, we show that orientation-aware geometric regularizers (Structure Tensor ST-DIP at 25.06 dB and Euler's Elastica at 24.87 dB) alongside L1/TV objectives selectively eliminate late-stage null-space checkerboards and contour wiggles, outperforming generic isotropic denoisers like DeepRED. Finally, we demonstrate that architectural generator priors vastly outperform internal patch recurrence (ZSSR and single-image DDPM), and confirm that the fundamental 'noise impedance' property fitting coherent natural structures faster than noise generalizes beyond CNNs to Vision Transformers.

Problem Formulation

The goal of super-resolution is to take a low resolution (LR) image x0 ∈ R3×H×W and upsampling factor t, and generate a corresponding high resolution (HR) version x ∈ R3×tH×tW .

Upsampling factor t is something that's needed to be chosen prehand.
We use t=4 in the whole of the experiments

Thus, Super-Resolution (SR) aims to invert the forward degradation process:

$$y = A(X) + \eta$$

where:

- $X \in \mathbb{R}^{3\times R\times H}$ is the unknown ground truth High-Resolution (HR) image.
- $A(\cdot)$ is the 4x spatial downsampling operator (Lanczos-2 kernel with anti-aliasing).
- $y \in \mathbb{R}^{3\times R/4\times H/4}$ is the observed Low-Resolution (LR) input image.
- η is measurement noise.

Strict Real-World Constraint: Zero-Shot / Single-Image Learning

In practical scenarios, external high-resolution training datasets are unavailable or unrepresentative of real camera degradations. Therefore, all methods evaluated here strictly adhere to **single-image unsupervised learning**:

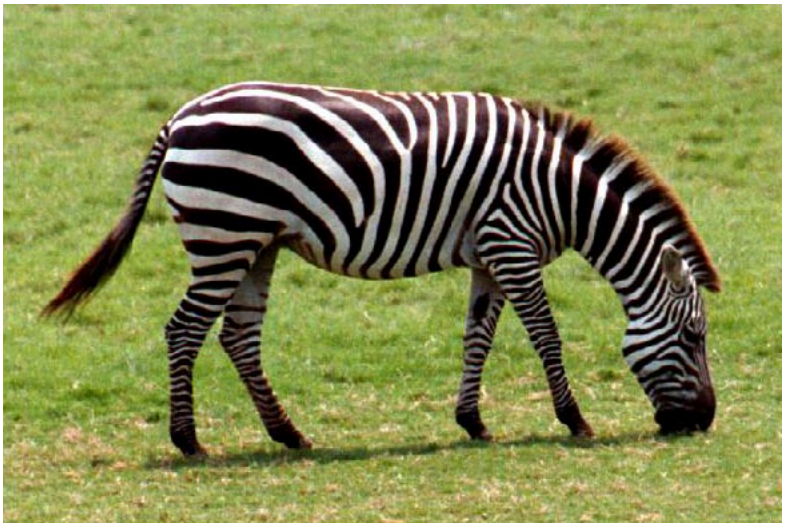
> **The ground-truth HR image X is NEVER exposed during training.**
> Any learning or optimization is driven solely by the input observation y and the degradation operator A.

Evaluation Strategy

Test Data

DIP [1] uses Set5 [2] and Set14 [3] datasets. We constrain ourselves to Set14 dataset. Most of the inital experiments have been carried out only on a single image of Set 14.

Original HR Image of R*3×576×384



The dimentions of all images in Set 14 dataset are available

Metric

We use PSNR as the metric for comparision calculated with using checkpoint of model with best psnr during the whole run.(Average PSNR in the case of Set 14 Dataset) as followed by [1][4].

Models

Conventional Baseline

As used in DIP[1], Conventional Baseline is Bicubic Interpolation and Nearest Neighbor.

Deep Image Prior

Our baseline comparision is with DIP[1].
DIP uses an fully-convolutional architectures, where the input z ∈ RC′×W ×H has the same spatial resolution as the the output of the network fθ (z) ∈ R3×W ×H . It uses an hour glass architecture. The hyperparameter details are kept same as what's presented in the paper

z ∈ R32×W ×H ~ U (0, 0.1)
nu = nd = [128, 128, 128, 128, 128]
ku = kd = [3, 3, 3, 3, 3]
ns = [4, 4, 4, 4, 4]
ks = [1, 1, 1, 1, 1]
σp = 1
30
num iter = 2000
LR = 0.01
upsampling = bilinear

Note: Upsampling is not learned, it's kept as unlearned and as mentioned in DIP[1], the results obtained were worse.

Principle: Fits the parameters θ of an untrained skip convolutional generator $f_\theta(z)$ directly to the LR image y:

$$\min_\theta \|A(f_\theta(z)) - y\|_2^2$$

where $z \sim \mathcal{N}(0, 0.1 \cdot \mathbf{I})$ is a fixed spatial noise tensor.

For more details refer to DIP[1].

Changing the architecture

It was carried out to explore how different neural network architectures perform as the generative prior in the Deep Image Prior (DIP) framework for 4x Super-Resolution to investigates which architectural inductive bias works best when optimizing on a single image without any external dataset training.

Explored models are

- unet (models.unet.UNet): Classic U-Net encoder-decoder (~2.16M parameters).
- resnet (models.resnet.ResNet): Deep Residual Network prior with 10 residual blocks (~797k parameters).
- swin (models.swin.swin): Vision Transformer prior using Swin Transformer shifted window attention (~2.88M parameters).[7]
- swin_skip (models.swin_skip.swin_skip): Recursive Swin Transformer equipped with multi-scale skip connections (~4.68M parameters).
- Dpwise-346K,100K: DIP[1] Architecture with 3d conv replaced with depth wise seperable convolutions. (Note : 2500 iterations were used in this case)

DIP + Regularization by Denoising [4]

- **Principle**: Incorporates an explicit cross-scale denoiser $D(\cdot)$ as an engine of regularization (RED):

$$\mathcal{L}(\theta) = \frac{1}{2}\|Af_\theta(z) - y\|_2^2 + \frac{\lambda_{\text{RED}}}{2} f_\theta(z)^T \Big(f_\theta(z) - D(f_\theta(z))\Big) + \lambda_{\text{TV}}\text{TV}(f_\theta(z))$$

- **Parameters**: $\lambda_{\text{RED}} = 0.005$, $\lambda_{\text{TV}} = 10^{-8}$

Single-Image Internal Learning (ZSSR-style UNet) [5]

- **Principle**:
  This approach trains an image-specific neural network exclusively on the internal cross-scale self-similarity of the single input image y.
  Leverages internal cross-scale recurrence (self-similarity across scales within a natural image).
- **Internal Dataset Construction (from y alone)**:
  i. Extract random crops $P_{\text{hr_sub}} \subset y$ of sizes 48×48.
  ii. Apply data augmentations: 4 discrete rotation angles ($0^\circ, 90^\circ, 180^\circ, 270^\circ$) and random horizontal/vertical flips.
  iii. Downsample each augmented crop using the degradation operator A to yield $P_{\text{lr_sub}} \in \mathbb{R}^{3\times12\times12}$.
  iv. Yields a self-supervised internal training dataset of 256 pairs $(P_{\text{lr_sub}}, P_{\text{hr_sub}})$.
- **Architecture**: Residual UNet predicting high-frequency residual Δ:

$$\hat{P}_{\text{hr}} = \text{Bicubic}(P_{\text{lr_sub}}) + \Delta_\theta(\text{Bicubic}(P_{\text{lr_sub}}))$$

- **Objective**: Charbonnier smooth L1 loss + Total Variation penalty:

$$\mathcal{L} = \sqrt{\|\hat{P}_{\text{hr}} - P_{\text{hr_sub}}\|^2 + \epsilon^2} + \lambda_{\text{TV}}\text{TV}(\hat{P}_{\text{hr}})$$

- **Test-Time Inference & Back-Projection**:
  After training terminates on the patch dataset, the **entire full image** y is passed through the network:

$$\hat{X}^{(0)} = \text{Model}(y)$$

  Followed by iterative back-projection data consistency:

$$\hat{X}^{(k+1)} = \hat{X}^{(k)} + \mu \cdot \text{Bicubic}\Big(y - A(\hat{X}^{(k)})\Big)$$

Single-Image Conditional Diffusion Model (Score-Based SR) Inspired by DDIM [6]

1. Why Diffusion / Score-Based Modeling?

Traditional regression models suffer from the "regression to the mean" problem (blurring uncertain high frequencies). A conditional score-based diffusion model generates sharp, realistic edge distributions conditioned on the low-resolution image.

2. Noise Schedule & Forward Process

- **Total Timesteps**: $T = 300$.
- **Beta Schedule**: Linear variance schedule $\beta_t \in [10^{-4}, 0.02]$.
- **Cumulative Alphas**:

$$\alpha_t = 1 - \beta_t, \quad \bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$$

- **Forward Noising Equation**:

$$q(x_t \mid x_0) = \mathcal{N}\Big(\sqrt{\bar{\alpha}_t} x_0, (1 - \bar{\alpha}_t)\mathbf{I}\Big) \implies x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon, \quad \epsilon \sim \mathcal{N}(0, \mathbf{I})$$

3. Training Architecture & Objective

- **Conditioning**: $c = \mathrm{Bicubic}(P_{\mathrm{lr}})$, upsampled to match the resolution of P_{hr}.
- **Score Network**: `ConditionalDiffusionUNet` with sinusoidal time embeddings:

$$\epsilon_\theta(x_t, t, c)$$

- **Training Loss**: Denoising Score Matching:

$$\mathcal{L}_{\mathrm{diff}}(\theta) = \mathbb{E}_{t, x_0, \epsilon}\Big[\|\epsilon_\theta(x_t, t, c) - \epsilon\|_2^2\Big]$$

  Randomly samples $t \sim \mathcal{U}\{0, T-1\}$, adds noise to internal patch $x_0 = P_{\mathrm{hr}}$, and trains ϵ_θ to predict the injected noise.

4. Inference & Diffusion Posterior Sampling (DPS)

- **Reverse Trajectory**: Subsampled $S = 50$ DDIM-style steps spanning $t = 299 \to 0$.
- **Starting Point**: Pure Gaussian noise $x_T \sim \mathcal{N}(0, \mathbf{I})$ at full target resolution (576×384).
- **Conditioning Input**: $c = \mathrm{Bicubic}(y)$.
- **Measurement Consistency Projection**:
  At each reverse step t, the predicted clean image $\hat{x}_0$ is projected toward the measurement constraint:

$$\hat{x}_0 = \frac{x_t - \sqrt{1 - \bar{\alpha}_t}\epsilon_\theta(x_t, t, c)}{\sqrt{\bar{\alpha}_t}}$$

$$\hat{x}_0 \leftarrow \mathrm{clamp}\Big(\hat{x}_0 + \gamma \cdot \mathrm{Bicubic}\big(y - A(\hat{x}_0)\big), 0, 1\Big)$$

  guaranteeing that the generated super-resolution output strictly downsamples back to the observed input y.

Changing the Regularization

Edge-Guided Null-Space & Contour DIP (EGNC-DIP)

A Novel Super-Resolution-Specific Regularization Framework for Deep Image Prior

Motivation and SR-Specific Physics

In super-resolution (SR), the forward degradation operator $A : \mathrm{HR} \to \mathrm{LR}$ maps a high-resolution (HR) image to a low-resolution (LR) image. Its null space is defined as

$$\mathcal{N}(A) = \{h : A(h) \approx 0\}.$$

Any high-frequency artifact h in the null space incurs approximately zero additional data loss:

$$\|A(f_\theta(z) + h) - y\|^2 \approx \|A(f_\theta(z)) - y\|^2.$$

Vanilla Deep Image Prior (DIP) can introduce spurious high-frequency noise and checkerboard artifacts in this null space during late optimization (typically beyond 3,000 steps).

Proposed SR-Specific Regularizer

1. **Flat-Region Null-Space Suppression**

In flat regions of the LR image, where the gradient magnitude of the upsampled image is small, high-frequency content that is unsupported by the observations may correspond to spurious noise.

We penalize deviations from the bicubic reconstruction:

$$R_{\text{flat}} = \|(1 - M_{\text{edge}}) \odot (f_\theta(z) - \text{Bicubic}(A(f_\theta(z))))\|_2^2 .$$

Here, M_{edge} is an edge mask, and $\odot$ denotes element-wise multiplication.

2. **Edge-Contour Tangential Continuity**

In natural images, true edges—such as zebra stripes—often form locally continuous contours with relatively small intensity variation along the tangent direction. Given the image gradient $\nabla f_\theta(z) = (f_x, f_y)$, define the tangent direction as

$$t = (-f_y, f_x).$$

We penalize variations along the tangent direction within edge regions:

$$R_{\text{contour}} = \|M_{\text{edge}} \odot (\nabla f_\theta(z) \cdot t)\|_1 .$$

This regularizer encourages smooth, continuous edge contours while preserving sharp transitions across edges and suppressing jagged artifacts.

Changing the Loss Function

L1

Replacing L2 Loss with L1 Loss

TV

L2 Loss complemented with TV Loss

Structure Tensor Coherence Regularized DIP (ST-DIP)

A novel Super-Resolution loss functin that exploits the eigensystem of the local Structure Tensor
to eliminate multi-directional high-frequency noise and checkerboard artifacts without blurring edges.

- **Mathematical Principle**:
  Analyzes the local 2D gradient orientation via the smoothed structure tensor:

$$J_\rho = K_\rho * \begin{bmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{bmatrix}$$

  where the eigenvalues $\mu_1 \geq \mu_2 \geq 0$ define the local geometry:
  - μ_1: Gradient energy along the dominant edge normal.
  - μ_2: Spurious gradient energy perpendicular to the dominant orientation (isotropic noise, null-space wiggles, checkerboards).
  - Local Anisotropy / Coherence: $C = \left(\frac{\mu_1 - \mu_2}{\mu_1 + \mu_2 + \epsilon}\right)^2$.
  - Regularization Penalty:

$$\mathcal{L}_{\text{ST}} = \frac{1}{|\Omega|} \sum_{(x,y)} \mu_2(x, y)$$

- **Why ST-DIP is Uniquely Suited for SR**:
  Natural edges (e.g., zebra stripes) have rank-1 structure tensors ($\mu_1 \gg 0, \mu_2 \approx 0$). In contrast, DIP null-space noise and checkerboard artifacts produce non-zero μ_2 everywhere. Minimizing μ_2 specifically purges non-directional noise without blunting stripe contrast!

Wavelet High-Frequency Sparsity & Directional Cross-Band DIP (Wavelet-DIP)

A novel Super-Resolution loss function that operates in the discrete wavelet transform (DWT) domain.

- **Mathematical Principle**:
  Applies an orthonormal 2D Discrete Wavelet Transform (DWT) decomposing $\hat{X}$ into subbands:

$$[LL, LH, HL, HH] = \mathrm{DWT}_{2D}(\hat{X})$$

  i. **Besov Sparsity Prior**:

$$\mathcal{L}_{\text{sparse}} = \|LH\|_1 + \|HL\|_1 + 2 \cdot \|HH\|_1$$

  ii. **Directional Cross-Band Mutual Exclusivity**:
  Physical stripes are purely horizontal ($LH \gg 0, HL \approx 0$) or purely vertical ($HL \gg 0, LH \approx 0$). Spurious DIP checkerboard artifacts create simultaneous energy in both ($|LH| \cdot |HL| \gg 0$).

$$\mathcal{L}_{\text{cross}} = \| |LH| \odot |HL| \|_1$$

Euler's Elastica Level-Line Curvature Regularized DIP (Elastica-DIP)

A novel Super-Resolution loss function rooted in differential geometry and variational mechanics.

- **Mathematical Principle**:
  Rooted in Euler (1744) and Mumford-Shah variational geometry:

$$\mathcal{L}_{\text{elastica}} = \int_{\Omega} (\alpha + \beta \kappa^2) |\nabla I|_{\epsilon} \, dx dy$$

  where $\kappa = \mathrm{div}\left(\frac{\nabla I}{|\nabla I|_{\epsilon}}\right)$ is the curvature of the image level lines (isophotes).
  - Penalizes level-line bending energy ($\kappa^2 |\nabla I|_{\epsilon}$), eliminating high-frequency oscillations while keeping smooth stripe contours perfectly intact.

Physical Justification for Super-Resolution:

1. High-frequency null-space noise and checkerboard artifacts cause rapid spatial oscillations
   in the isophote orientation, causing \kappa^2 to explode.
2. True physical edges (e.g. zebra stripes, boundaries) have smooth, coherent level-line trajectories
   with near-zero curvature (\kappa \approx 0).
3. By penalizing level-line bending energy (\kappa^2 |\nabla I|), Elastica selectively extinguishes
   chaotic null-space fluctuations while completely preserving sharp, low-curvature stripe edges!

Spectral Power-Law Decay DIP (Spectral-DIP)

A novel Super-Resolution loss function operating in the 2D Fourier Frequency domain.

- **Mathematical Principle**:
  Natural images adhere to Field's (1987) universal power law:

$$S(r) \propto r^{-2\alpha}, \quad \alpha \approx 1$$

  where $S(r)$ is the radially averaged Fourier power spectrum at spatial frequency radius r.
  For spatial frequencies in the null-space $r > r_{\text{cutoff}} = 1/(2 \times \text{factor})$, the downsampling operator provides 0 supervision.
  Spectral-DIP penalizes any high-frequency spectral violation where power fails to decay at the physical r^{-2} rate:

$$\mathcal{L}_{\text{spectral}} = \sum_{k \geq k_{\text{cutoff}}} \mathrm{ReLU}\left(\log E_{k+1} - \log E_k + 2\log(r_{k+1}/r_k)\right)^2$$

Architecture + Loss Function

Dpwise-346K,100K-DeepRED: DIP[1] Architecture with 3d conv replaced with depth wise seperable convolutions and using DeepRED Loss function. (Note : 2500 iterations were used in this case)

Training Configuration

Since one of the major concerns of DIP[1] is that we don't known when to stop. All the experiments apart few (mentioned specifically) used 6000 iterations

Optimizer and learning rate was kept same as DIP[1] which is ADAM optimizer.

Results on Single Image

6000 Steps

What changed?	Method	Parameters	PSNR vs Real HR (dB)	Gain vs Bicubic (dB)	Gain vs Vanilla DIP (dB)	Artifact Directory	W&B Run
Baseline	Nearest Neighbor	0	20.82 dB	-2.28	-3.92	output_dip_sr/	-
Baseline	Bicubic Interpolation	0	23.10 dB	0	-1.64	output_dip_sr/	-
Baseline	Vanilla DIP (Skip UNet)	2217831	24.74 dB	1.64	0	output_dip_sr/	dip_sr_zebra_x4_seed_0
Regularization	DIP + Regularization by Denoising (Deep Red)		24.81 dB	1.71	0.07	output_dip_red/	dip_red_sr_zebra_x4_seed
Loss Function	L1		25.08	1.98	0.34	output_dip_losses_sr	skip_losses_zebra
Loss Function	TV		25.14	2.04	0.4	output_dip_losses_sr	skip_losses_zebra
Loss Function	Internal ZSSR UNet		20.96 dB	-2.14	-3.78	output_single_image_internal_sr/	zssr_internal_sr_zebra_x4_
Loss Function	Single-Image Conditional DDPM		20.66 dB	-2.44	-4.08	output_single_image_ddpm_sr/	single_image_ddpm_sr_zeb
Loss Function	Edge-Guided Null-Space & Contour DIP (EGNC-DIP)		24.88 dB	1.78	0.14	output_dip_contour_sr/	dip_egnc_sr_zebra_x4_see
Loss Function	Structure Tensor Coherence DIP (ST-DIP)		25.06 dB	1.96	0.32	output_dip_structure_tensor/	dip_st_sr_zebra_x4_seed_(
Loss Function	Wavelet Sparsity & Directional DIP (Wavelet-DIP)		23.64 dB	0.54	-1.1	output_dip_wavelet/	dip_wavelet_sr_zebra_x4_s
Loss Function	Euler's Elastica Curvature DIP (Elastica-DIP)		24.87 dB	1.77	0.13	output_dip_elastica/	dip_elastica_sr_zebra_x4_s
Loss Function	Spectral Power-Law Decay DIP (Spectral-DIP)		24.71 dB	1.61	-0.03	output_dip_spectral/	dip_spectral_sr_zebra_x4_s
Architecture	U-Net	2164899	24.09	0.99	-0.65	output_all_models_sr_1	all_models_sr_zebra_x4
Architecture	ResNet	797123	16.49	-6.61	-8.25	output_all_models_sr_2	all_models_sr_zebra_x5

What changed?	Method	Parameters	PSNR vs Real HR (dB)	Gain vs Bicubic (dB)	Gain vs Vanilla DIP (dB)	Artifact Directory	W&B Run
Architecture	Swin Transformer	2887303	23.06	-0.04	-1.68	output_all_models_sr_3	all_models_sr_zebra_x6
Architecture	Swin + Skip	4687663	22.62	-0.48	-2.12	output_all_models_sr_4	all_models_sr_zebra_x7

2500 Steps

Experiment	What changed?	Method	Parameters	PSNR vs Real HR (dB)	Gain vs Bicubic (dB)	Gain vs Vanilla DIP (dB)	Artifact Directory	W&B Run
Architecture	Architecture	Dpwise-346K	346171	23.88	0.78	-0.86	output_zebra_scaling/	dip-zebra-scaling
Architecture + Loss Function	Architecture + Loss Function	Dpwise-346K-DeepRED	346171	24.04	0.94	-0.7	output_zebra_scaling/	dip-zebra-scaling
Architecture	Architecture	Dpwise-1M	1013851	24.59	1.49	-0.15	output_zebra_scaling/	dip-zebra-scaling
Architecture + Loss Function	Architecture + Loss Function	Dpwise-1M-DeepRED	1013851	24.56	1.46	-0.18	output_zebra_scaling/	dip-zebra-scaling

4000 Iterations

Experiment	What changed?	Method	Parameters	PSNR vs Real HR (dB)	Gain vs Bicubic (dB)	Gain vs Vanilla DIP (dB)	Artifact Directory	W&B Run
Architecture	Architecture	Convnext		24.9	1.8	0.16	output_convnext_multi_image	dip-convnext-multi
Architecture + Loss Function	Architecture + Loss Function	Convnext + Red Loss Function		24.9	1.8	0.16	output_convnext_multi_image	dip-convnext-multi

Results on Set 14

After running for 4000 iterations.
Artifact Directory output_convnext_multi_image, output_multi_image_sr

Image	Bicubic (dB)	Skip-L2 (DIP) (dB)	Skip-DeepRED (dB)	DPwise-L2 (dB)	DPwise-DeepRED (dB)	ConvNeXt-L2 (dB)	ConvNeXt-RED (dB)
img_001	20.31	20.44 (+0.13)	20.45 (+0.14)	20.29 (−0.02)	20.23 (−0.08)	20.51 (+0.20)	20.52 (+0.21)
img_002	23.58	23.91 (+0.33)	23.91 (+0.33)	23.53 (−0.05)	23.51 (−0.07)	23.94 (+0.36)	23.95 (+0.37)
img_004	24.05	24.55 (+0.50)	24.60 (+0.55)	24.57 (+0.52)	24.54 (+0.49)	24.61 (+0.56)	24.65 (+0.60)
img_005	20.27	21.14 (+0.87)	21.17 (+0.90)	20.93 (+0.66)	20.90 (+0.63)	21.04 (+0.77)	21.00 (+0.73)

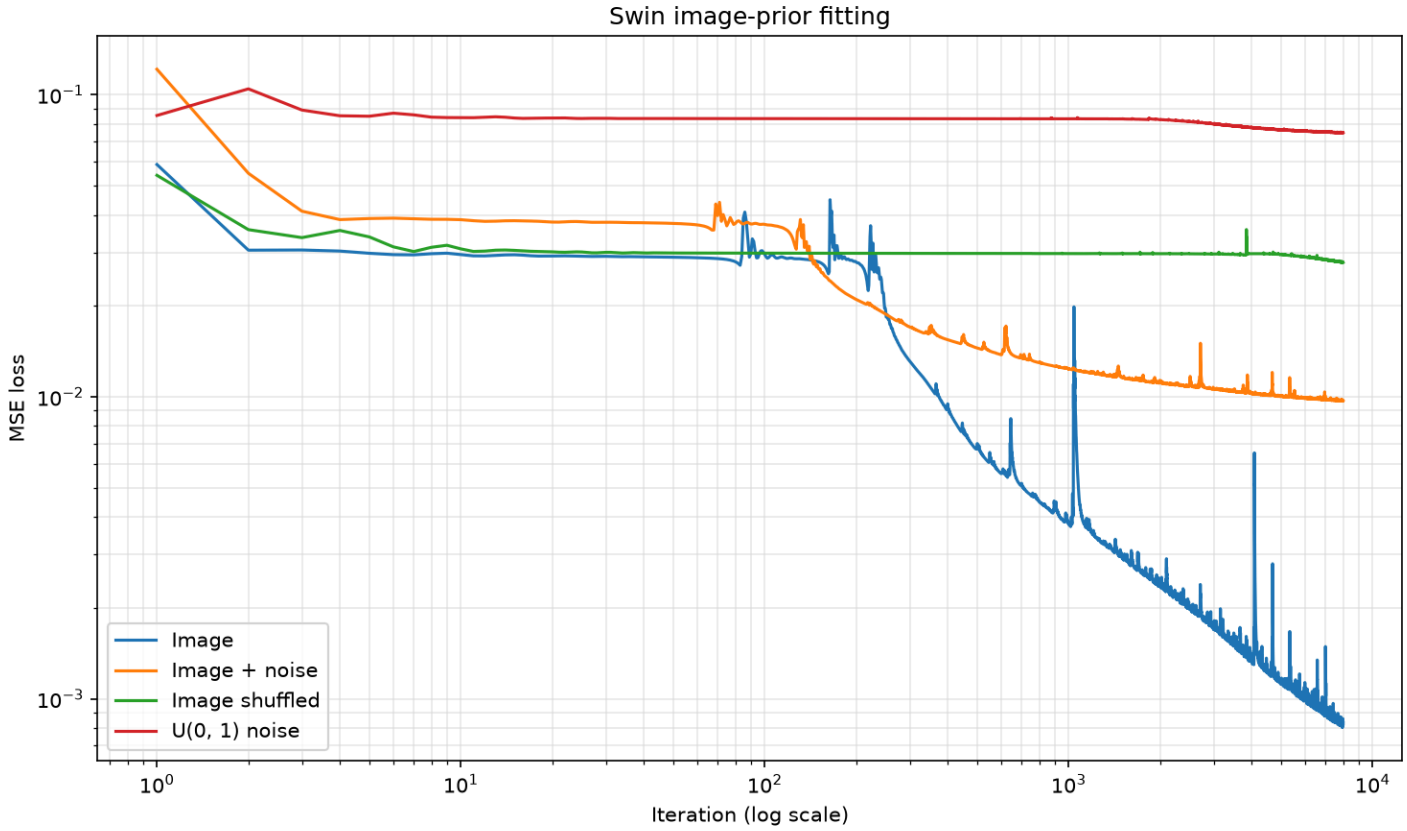
Image	Bicubic (dB)	Skip-L2 (DIP) (dB)	Skip-DeepRED (dB)	DPwise-L2 (dB)	DPwise-DeepRED (dB)	ConvNeXt-L2 (dB)	ConvNeXt-RED (dB)
img_006	28.83	29.13 (+0.30)	29.10 (+0.27)	29.03 (+0.20)	29.05 (+0.22)	29.23 (+0.40)	29.29 (+0.46)
img_007	23.78	25.05 (+1.27)	25.07 (+1.29)	24.74 (+0.96)	24.77 (+0.99)	24.97 (+1.19)	24.86 (+1.08)
img_008	26.12	28.19 (+2.07)	28.30 (+2.18)	27.61 (+1.49)	27.07 (+0.95)	28.24 (+2.12)	28.42 (+2.30)
img_009	28.56	29.52 (+0.96)	29.54 (+0.98)	29.14 (+0.58)	29.07 (+0.51)	29.52 (+0.96)	29.57 (+1.01)
img_011	26.54	29.35 (+2.81)	29.29 (+2.75)	28.56 (+2.02)	28.79 (+2.25)	29.15 (+2.61)	29.04 (+2.50)
img_012	27.34	28.56 (+1.22)	28.40 (+1.06)	27.51 (+0.17)	27.52 (+0.18)	28.65 (+1.31)	28.62 (+1.28)
img_013	20.31	23.24 (+2.93)	23.37 (+3.06)	22.96 (+2.65)	22.88 (+2.57)	23.41 (+3.10)	23.51 (+3.20)
img_014	23.1	24.94 (+1.84)	25.01 (+1.91)	24.49 (+1.39)	24.52 (+1.42)	24.90 (+1.80)	24.90 (+1.80)
Average	24.4	25.67	25.68	25.28	25.24	25.69	25.69

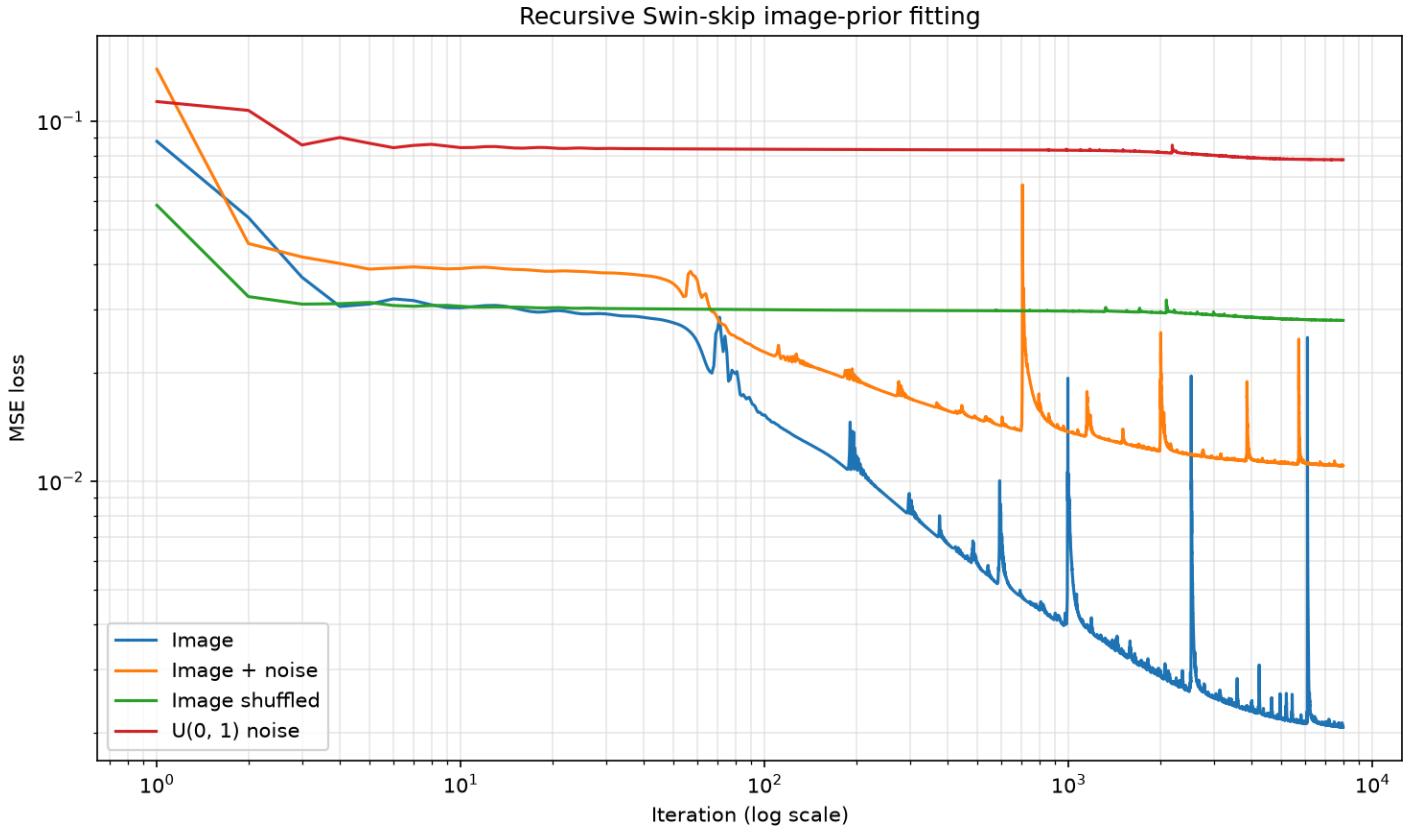
Experiment: Testing of Noise Impedence Property

The most intersting part is test of noise impedence property which shows that architecture inherently inhibits bias about images.

We tested with SWIN Transformer [7] and it's variant by adding similar to DIP multi-scale skip connections. Explicit 1x1 Conv + BatchNorm + LeakyReLU skip pathways at each scale, matching classic DIP.In terms of parameter size it was 1.6x larger than the former. This was designed to test whether giving the Swin Transformer the exact multi-scale skip inductive bias of Ulyanov's convolutional DIP network would improve its natural image prior.

hyperparameters used,
STEPS = 8000
INPUT_DEPTH = 32
PATCH_SIZE = 4
EMBED_DIM = 32
SWIN_DEPTH = 2
NUM_HEADS = 4
WINDOW_SIZE = 8
LEARNING_RATE = 0.0001
NOISE_SIGMA = 25.0 / 255.0





The test of noise impedance almost holds the same as the test shown in the paper. So the initial part is not quite in the same order, but at the end, the order remains the same. This was a Transformer. So this also shows that it is not just an inherent property of CNN; it is also an inherent property of Transformers too. But Transformers have no locality bias. The fact that it is still able to do that is still a very big thing. And so, there is no deep doubt on why such noise impedance property gets exhibited, since which is still unknown. This was not expected for the transformer.

Key Research Findings & Takeaways:

1. **Topology & Multi-Scale Hierarchy Over Parameter Volume (The Depthwise-Separable Discovery)**:

- Slashing the parameter count by **84%** (from 2,217,831 down to 346,171) by replacing standard 3x3 convolutions with Depthwise-Separable Convolutions (`dpwise_skip`) causes negligible performance loss.
- Across the 12-image Set14 benchmark, `DPwise-L2` achieves an average of **25.28 dB** compared to full DIP's **25.67 dB** (+0.88 dB over Bicubic), preserving over **70% of DIP's gain** with only 1/6th of the parameter footprint.
- Scaling depthwise width to 1.01M parameters (`width=224`) closes the gap on zebra to within **0.15 dB** of the full 2.2M baseline (24.59 dB vs 24.74 dB).
- **Core Takeaway**: DIP's implicit natural image prior is primarily governed by the **multi-scale Hourglass/U-Net topology and spatial downsampling/upsampling pyramid**, NOT by dense channel-to-channel weight volume. The network can be drastically compressed for real-time single-image inverse problems without collapsing the image prior.

2. **Architectural Inductive Hierarchy: Modern ConvNeXt >~ Skip UNet > Vision Transformers >> Flat ResNets**:

On zerbra image, Modern ConvNeXt achieves 24.9
, DIP achieves 24.74 dB, Swin transformer 23.06 and 22.62 with skip connections and resnet achieves 16.49 db.

- **Vision Transformers (Swin & Swin+Skip)** lag significantly behind CNNs (~22.6–23.0 dB, ~1.7–2.1 dB below Skip). Despite self-attention and up to 4.69M parameters, Transformers lack the built-in local translational equivariance and multi-scale receptive field hierarchy that make convolutional networks converge rapidly on single images.
- **Flat ResNet completely fails (16.49 dB, -6.61 dB below Bicubic)**: Without multi-scale down/upsampling pyramids, flat residual blocks cannot translate random spatial noise into multi-frequency natural image statistics.

Note: Though the experiment has been conducted only for one image

ConNext and Skip Unet (DIP) achieves same average PSNR on 12 images. So no method is superior.

3. **Loss Function Geometry: Orientation-Aware & L1/TV Losses Outperform Generic Denoising**:

- Switching from L2/MSE to **Total Variation (25.14 dB)** or **L1 Loss (25.08 dB)** yields an immediate +0.34 to +0.40 dB boost over Vanilla DIP because L1/TV penalties do not over-penalize sharp step discontinuities (edges).
- **Structure Tensor Coherence (ST-DIP, 25.06 dB)** and **Euler's Elastica (Elastica-DIP, 24.87 dB)** selectively eliminate spurious null-space checkerboards and high-curvature isophote wiggles without blunting directional stripe edges.
- In contrast, generic bilateral denoisers (**DeepRED, 24.81 dB**) and transform sparsity (**Wavelet-DIP, 23.64 dB**) hit a ceiling because isotropic smoothing blunts fine directional textures.

Note: Though the experiment has been conducted only for one image

4. **Limits of Internal Self-Similarity (ZSSR & DDPM) vs. Architectural Prior**:

- Single-image patch recurrence models (ZSSR UNet at 20.96 dB and DDPM SR at 20.66 dB) severely underperform DIP (24.74+ dB).
- **Why**: Extracting patches from an already degraded LR input heavily attenuates the high-frequency spectrum. The networks learn the degraded patch distribution accurately, but lack the structural inductive prior necessary to extrapolate unseen high frequencies. An untrained multi-scale generator network (DIP) is far more effective as a single-image prior than cross-scale patch recurrence.
  Note: Though the experiment has been conducted only for one image

5. **Noise Impedance is an Inherent Deep Network Property, Not Just CNN Locality**:
   Noise Impedence Experiment
   - Experiments on the Swin Transformer confirm that the noise impedance property (fitting natural images orders of magnitude faster than scrambled or pure Gaussian noise) holds for self-attention architectures as well.
   - This demonstrates that spectral bias toward low-entropy, coherent visual structures is a universal property of deep parameterized models, rather than an artifact exclusive to convolutional weight sharing.
   - Though the time of converagance is much higher for transformer as comapred to CNN but it shouldn't effect the analysis of noise impedence property. (Best to our knowledge)

Limitations

- Most of the analysis was conducted on a single image thus the results might not be statistically significant
- Only Quantitative Analysis was carried out. Qualitative Analysis might be important part that could add more analysis and justify the super resolution task in a better way.
- For overall results on Set 14 dataset, we weren't able to find a method that outperforms DIP.
- No hyperparameter tuning was performed for any kind.
- The diffusion experiment, adding attention experiment and some of loss function experiments such as spectral loss, elastica, structured tensor were completely outsourced to Gemini, to check if we might get some meaningful results which could be explain.
- The learning rate tuning should have been done from the start, but we strongly believe the results we achieved will hold.

Conclusion

Most crucial thing that we are able to experimentally prove is that noise impedance is an inherent deep network property, not just CNN Locality.

Secondly, showing that image fitting is independent of the number of parameters rather it depends on architecture as evident in depth wise seperable experiments.

Addition of a good regularizer i.e specific to the task would always help in improving the performance, such as addition of RED and structued tensor regularization.

We experimented by changing the architecture, loss function and combination of both of them, which also includes experimenting novel loss funtions. Though we weren't able to achieve any significant improvement over DIP which was also limited by experimental approach we did on just 1 image, but overall the findings are beneficial for in pursuit of producing a possible better method.

The key takeaway from our experiments is, though if we might not be able beat DIP by some margin, we can always bargain a small amount of performance for fraction of parameters. (Depthwise + pointwise, ConvNext).

Addition of attention wouldn't help in any sense, since the most important part of the architecture is locality itself. (local similarities win over global similarities).

Future Work

- For the configuration with most promising results and also easy to understand which we ConvNext with L2 loss, we aim to do hyperparameter tuning.

Decleration of AI use

AI particularly Gemini has been used in writing the code for the experiments. DIP[1] code is borrowed from the authors themselves using exactly as it is.

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Appendix

Set 14 Image Sizes

Image File	Benchmark Name	Width × Height	Shape (H × W × C)
img_001_SRF_4_HR.png	baboon	500 × 480	480 × 500 × 3
img_002_SRF_4_HR.png	barbara	720 × 576	576 × 720 × 3
img_004_SRF_4_HR.png	coastguard	352 × 288	288 × 352 × 3
img_005_SRF_4_HR.png	comic	248 × 360	360 × 248 × 3
img_006_SRF_4_HR.png	face	276 × 276	276 × 276 × 3
img_007_SRF_4_HR.png	flowers	500 × 360	360 × 500 × 3

Image File	Benchmark Name	Width × Height	Shape (H × W × C)
img_008_SRF_4_HR.png	foreman	352 × 288	288 × 352 × 3
img_009_SRF_4_HR.png	lenna	512 × 512	512 × 512 × 3
img_011_SRF_4_HR.png	monarch	768 × 512	512 × 768 × 3
img_012_SRF_4_HR.png	pepper	512 × 512	512 × 512 × 3
img_013_SRF_4_HR.png	ppt3	528 × 656	656 × 528 × 3
img_014_SRF_4_HR.png	zebra	584 × 388	388 × 584 × 3

Note: Image 3 and Image 10 is missing from the dataset